# Software Engineering Practices in Midas

**Date:** 2025-09-07

**Version Context:** v0.3.7-cleanup

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## 1. Version Control

- **Practice:** Frequent commits and annotated Git tags (v0.3.4, v0.3.5, v0.3.7-cleanup).
- **Software Principle:** Versioning and release management.
- **Impact:** Provides restore points, traceability, and confidence in experimentation.

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## 2. Safe Archiving

- **Practice:** Files are moved to `_archive_unused/YYYYMMDD/` instead of deleted.
- **Software Principle:** Reversibility and audit trail.
- **Impact:** Maintains a clean repo without permanent loss of history.

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## 3. Documentation Discipline

- **Practice:** MD + PDF files for each milestone (sanity checks, workflow principles, audit reports).
- **Software Principle:** Knowledge capture and project traceability.
- **Impact:** Enables future maintainers (or your future self) to understand “why” as well as “what.”

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## 4. Modularity in Scenarios

- **Practice:** Strategies separated into A–E scenarios. Each can be run, compared, and tuned independently.
- **Software Principle:** Separation of concerns and modular design.
- **Impact:** Simplifies debugging, testing, and later hybridization.

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## 5. Guardrails Before Enhancements

- **Practice:** Prioritize EMA/VWAP confirms, MACD rise bars, RVOL at open, universe hygiene.

- **Software Principle:** Build correctness and stability before optimization.

- **Impact:** Ensures stability before pursuing advanced features like adaptive sizing or hybrid routers.

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## 6. Future-Proof Planning

- **Practice:** Broker abstraction, adaptive sizing, hybrid router deferred until baseline stable.

- **Software Principle:** Incremental development and roadmap-driven design.

- **Impact:** Prevents premature complexity and protects maintainability.

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## 7. Automation Preference

- **Practice:** Defaulting to Python for runners and utilities, avoiding brittle PowerShell scripts.

- **Software Principle:** Reliability, portability, and minimizing tech debt.

- **Impact:** Reduces errors and aligns with professional engineering best practice.

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## Conclusion

The Midas project demonstrates **professional software engineering discipline**:

- Frequent versioning

- Safe cleanup and reversibility

- Documentation and traceability

- Modular scenario design

- Guardrails before profit

- Deferred complexity until stability

Even if not from a formal software career, the practices align closely with what strong engineers and architects use daily.